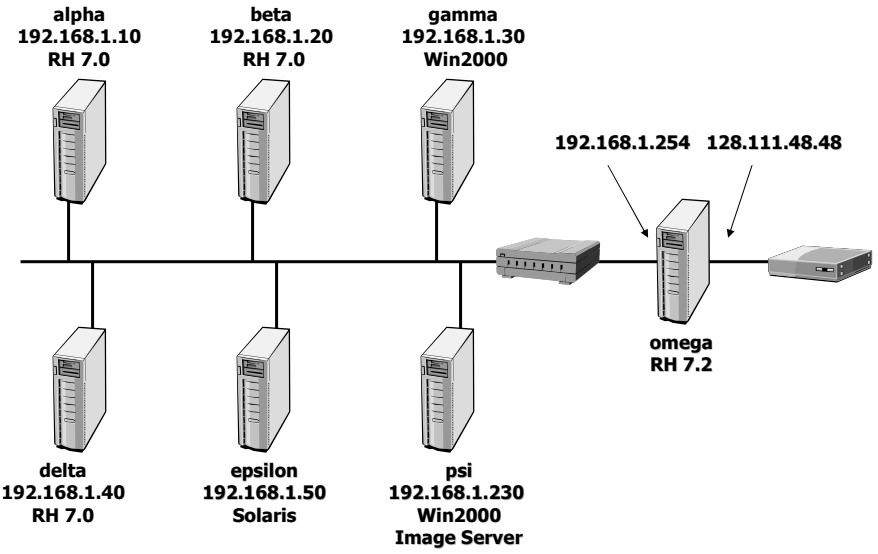
The Testbed Network

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Structure

- Testbed IP address are allocated in subnet 192.168.1
- Flat topology: hosts are connected by Ethernet hub
- Default gateway 192.168.1.254
- Currently available hosts have Linux Redhat 7.0 and Windows 2000 Server installed
- An additional host with Solaris will be available soon
- Hosts operating systems are re-initialized to a known state (empty home directories!!) every once in a while
  - In case of an unrecoverable error (whenever it happens)
  - Every once in a while (possibly early morining or late at night)

Topology



Access to the Testbed

- Testbed hosts can be accessed from *anywhere* through Secure Shell (ssh)
- Incoming connection to port 30-38 are mapped to addresses 192.168.1.10-192.168.1.100
- Firewall host performs Network Address Translation
- Outgoing traffic is not allowed to 128.111.48 network
- Initially available hosts: alpha, beta, gamma, delta
- Omega and psi are special hosts: **DO NOT TOUCH!!**

Access to the Testbed

- DO NOT USE TELNET FROM/TO THE OUTSIDE
- DO NOT USE FTP FROM/TO THE OUTSIDE
- Read and understand the SSH man page
- To transfer files from/to the outside use *always* Secure Copy (scp)
  - From the outside to the testbed
    % scp file handle@128.111.48.48: -p <the-right-port>
  - From the inside to the outside
    % scp handle@testbedhost:file . –p <the-right-port>
- Your CS password must never be used while on the testbed

Example

- Mark Twain (CS account: *mtwain,* handle: *bzero*) starts a session on alpha from host 128.111.48.69 (longboard.cs.ucsb.edu)
- ssh bzero@128.111.48.48 –p 30

  bzero@alpha's password:
  [bzero's password (not your CS password!)]

  Last login: Sun Apr 11 19:35:35 2002
  [...]

Example

- Moving file exploit.c to alpha

  longboard% scp exploit.c bzero@128.111.48.48: -p 30
  bzero@alpha's password:
  exploit.c | 1 KB | 0.0 kB/s | ETA: 00:00:00 | 100%
- Compiling the file

  alpha> gcc exploit.c -o exploit
- Running the program

  alpha> ./exploit > output.txt
- Moving output.txt back to longboard

  longboard% scp bzero@128.111.48.48:output.txt . –p 30
  bzero@alpha's password:
  output.txt | 2 KB | 0.0 kB/s | ETA: 00:00:00 | 100%

I WANT TO BE ROOT!

Acceptable Use

- The network is to be used for security testing only
  - It's not a storage space
  - It's not a development environment
- Resources are limited and there are many users
  - Try to limit your resource usage (disk, network, CPU)
- Be sure you know what you are doing
  - Do not execute programs whose behavior is unknown
  - Do not flood the network
    - Do not perform denial-of-service attacks for extended amounts of time
    - Do not start an attack if you don't know how to stop it!
  - Do not remove/access other users' files or change host configuration files

Acceptable Use

- Do not use the testbed to attack outside hosts
- Do not store anything important on the testbed
  - The hosts may go down at any moment
  - The network may go down at any moment
  - The disk may be formatted at any moment
- Please, try to leave the network as you found it
  - No trojan horses
  - No permanent side effects
  - No software installed in commonly used path (/bin, /usr/bin, /usr/local/bin, etc)
- Try to use your brains and have fun!

What If Something Goes Wrong?

- First try to understand what went wrong
  - Is it something related to your activities?
- Send email from your CS account
  - to vigna@cs
  - CC to kimaya@cs
  - CC to wmighao@cs
- Try to explain what happened
- Try to be clear and concise
  - YES: "host alpha does not respond to ping from host 192.168.1.20 [trace follows]"
  - NO: "The network doesn't work"

Caveats

- Expect a lot of frustration
- Do not keep important files on the testbed hosts
- Synchronize often
- You are many
- Be patient
- Check the class web page for updates